

Codes

A B C D	A B C D
(a) 1 2 4 3	(b) 3 1 5 4
(c) 5 2 1 4	(d) 3 1 2 4

➤ Previous Years' Questions

41. Which one among the following statements about an atom is not correct? ✎ 2012 I

- (a) Atom always combine to form molecules
- (b) Atoms are the basic units from which molecules and ions are formed
- (c) Atoms are always neutral in nature
- (d) Atoms aggregate in large numbers to form the matter that we can see, feel and touch

42. Which one among the following is not a correct statement? ✎ 2013 II

- (a) Cathode rays are negatively charged particles
- (b) Cathode rays are produced from all the gases
- (c) Electrons are basic constituents of all the atoms
- (d) Hydrogen ions do not contain any proton

43. In tritium (T), the number of protons (p) and neutrons (n) respectively, are ✎ 2014 II

- (a) 1 p and 1 n
- (b) 1 p and 2 n
- (c) 1 p and 3 n
- (d) 2 p and 1 n

44. The distribution of electrons into different orbits of an atom, as suggested by Bohr, is ✎ 2014 II

- (a) 2 electrons in the K -orbit, 6 electrons in the L -orbit, 18 electrons in the M -orbit
- (b) 2 electrons in the K -orbit, 8 electrons in the L -orbit, 32 electrons in the M -orbit
- (c) 2 electrons in the K -orbit, 8 electrons in the L -orbit, 18 electrons in the M -orbit
- (d) 2 electrons in the K -orbit, 8 electrons in the L -orbit, 16 electrons in the M -orbit

45. Who among the following proposed that atom is indivisible? ✎ 2015 I

- (a) Dalton
- (b) Berzelius
- (c) Rutherford
- (d) Avogadro

46. The atomic theory of matter was first proposed by ✎ 2015 II

- (a) John Dalton
- (b) Rutherford
- (c) J.J. Thomson
- (d) Niels Bohr

47. Which one of the following statements is not correct? ✎ 2016 II

- (a) Atoms of different elements may have same mass numbers
- (b) Atoms of an element may have different mass numbers
- (c) All the atoms of an element have same number of protons
- (d) All the atoms of an element will always have same number of neutrons

Radioactivity

48. In which categories did Marie Curie win her two different Nobel Prizes?

- (a) Physics and Chemistry
- (b) Chemistry and Medicine
- (c) Physics and Medicine
- (d) Chemistry and Peace

49. The α -particles are

- (a) high energy electrons
- (b) positively charged hydrogen ions
- (c) high energy X-ray radiations
- (d) double positively charged helium nuclei

50. Gamma rays have

- (a) zero mass and no charge
- (b) zero mass and positive charge
- (c) unit mass and zero charge
- (d) unit mass and negative charge

51. An α -particle consists of which of the following?

- (a) 2 protons and 2 neutrons
- (b) 1 proton and 1 electron
- (c) 2 protons and 4 neutrons
- (d) 1 proton and 1 neutron

52. Which of the following particles cannot be accelerated?

- (a) Electrons
- (b) α -particles
- (c) Neutrons
- (d) Protons

53. A nuclear reactor produce nuclear energy by

- (a) nuclear fusion
- (b) spontaneous fission
- (c) uncontrolled chain reaction
- (d) controlled chain reaction

54. In nuclear reactors, the speed of neutrons is slowed down by

- (a) heavy water
- (b) ordinary water
- (c) zinc rods
- (d) molten caustic soda

55. The energy generation in stars is (a) mainly due to the fusion of heavy nuclei

- (b) mainly due to the fusion of lighter nuclei
- (c) solely due to the fusion of heavy nuclei
- (d) due to both fission and fusion of lighter nuclei

56. The difference between a nuclear reactor and atomic bomb is that

- (a) no chain reaction takes place in nuclear reactor while in the atomic bomb, there is a chain reaction
- (b) the chain reaction in nuclear reactor is controlled
- (c) the chain reaction in nuclear reactor is uncontrolled
- (d) no chain reaction takes place in atomic bomb while it takes place in nuclear reactor

57. 'Yellow cake' an item of smuggling across the border is

- (a) a crude form of heroin
- (b) a crude form of cocaine
- (c) uranium oxide
- (d) unrefined gold

58. Hydrogen bomb is based on the principle of

- (a) nuclear fission
- (b) nuclear fusion
- (c) natural radioactivity
- (d) artificial radioactivity

59. In treatment of cancer, which of the following is used?

- (a) $^{131}_{53}\text{I}$
- (b) $^{32}_{15}\text{P}$
- (c) $^{60}_{27}\text{Co}$
- (d) ^2_1H

60. **Statement I** Hydrogen bomb is more powerful than atomic bomb.

Statement II In hydrogen bomb, reaction is initiated by fission followed by fusion.

Codes

- (a) Both the Statements are correct and Statement II is the correct explanation of Statement I
- (b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
- (c) Statement I is correct but Statement II is incorrect
- (d) Statement I is incorrect but Statement II is correct

61. Consider the following statements

1. Atomic number decreases by 2 and mass number increases by 4 on α -particle emission.
2. Atomic number increases by 1 and mass number remains unchanged on β -particle emission.

Choose the correct statement(s)

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) None of the above

62. Which one of the following is not needed in a nuclear fission reactor?

1. Moderator 2. Accelerator
3. Coolant

Choose the correct option.

(a) Only 1 (b) Only 2
(c) 1 and 2 (d) 2 and 3

63. Match the following Columns.

Column I (Nuclear reactor component)	Column II (Substance used)
A. Moderator	1. Uranium
B. Control rods	2. Graphite
C. Fuel rods	3. Boron
D. Coolant	4. Lead
	5. Sodium

Codes

A B C D
(a) 2 1 3 5
(b) 2 3 1 5
(c) 3 2 1 5
(d) 3 4 1 2

64. Consider the following statements In nuclear power reactors,

1. uranium-235 is used as the fission material.
2. graphite is used as a moderator.
3. rods of lead are used to control the rate of nuclear reaction.

Which of the above statements are correct?

(a) 1 and 3 (b) 1 and 2
(c) 2 and 3 (d) All of these

65. Consider the following statements

1. Gamma rays are not the constituents of nuclei but they are emitted when a nucleus in an excited state returns to its normal state.
2. The neutron to proton ratio for stable nuclei is always less than one.
3. Beta rays consist of ordinary electrons which are of nuclear origin but do not revolve in orbits.
4. X-rays and gamma rays are electromagnetic radiations.

Which of the above statements are correct?

(a) 2, 3 and 4
(b) 1, 3 and 4
(c) 1, 2 and 4
(d) 1, 2 and 3

66. Match the following Columns.

Column I (Radioisotopes)	Column II (Medicinal uses)
A. ^{60}Co	1. Leukemia
B. ^{131}I	2. Anaemia
C. ^{59}Fe	3. Cancerous tumours
D. ^{32}P	4. Disorders of thyroid gland

Codes

A B C D A B C D
(a) 3 4 1 2 (b) 1 2 3 4
(c) 3 4 2 1 (d) 4 3 2 1

67. Which one of the following is heavy water used in nuclear reactor?

(a) Water having molecular weight 18u
(b) Water having molecular weight 20u
(c) Water at 4°C but having molecular weight 19u
(d) Water below the ice in a frozen sea

68. The light emitted by firefly is due to

(a) the radioactive substance
(b) chemiluminescence process
(c) the photoelectric process
(d) burning of phosphorus

> Previous Years' Questions

69. Age of fossil may be found out by determining the ratio of two isotopes of carbon. The isotopes are

☞ 2012 I

(a) C-12 and C-13 (b) C-13 and C-14
(c) C-12 and C-14
(d) C-12 and carbon black

70. Which one of the following reactions is the main cause of the energy radiation from the sun?

☞ 2012 I

(a) Fusion reaction
(b) Fission reaction
(c) Chemical reaction
(d) Diffusion reaction

71. In an atomic explosion, release of large amount of energy is due to the conversion of

☞ 2012 II

(a) chemical energy into nuclear energy
(b) nuclear energy into heat
(c) mass into energy
(d) chemical energy into heat

72. Carbon or graphite rods are used in atomic reactors as moderators for sustained nuclear chain reaction through nuclear fission process. In this process, ☞ 2014 II

(a) the neutrons are made fast
(b) the protons are made fast

(c) the neutrons are made slow
(d) the protons are made slow

73. In an observation, α -particles, β -particles and γ -rays have same energies. Their penetrating power in a given medium in increasing order will be

☞ 2015 I

(a) α , β , γ (b) β , γ , α
(c) α , γ , β (d) β , α , γ

Chemical Bonding and Redox Reactions

74. In case of a cation,

(a) number of electrons are more than the number of protons
(b) number of electrons are equal to the number of protons
(c) number of electrons are more than the number of neutrons
(d) number of electrons are less than the number of protons

75. Atomic numbers of A and B are 2 and 12, respectively. Valency (or number of electrons taking part in bonding) is two in case of

(a) Only A (b) Only B
(c) Both A and B (d) Neither A nor B

76. A bond formed by the transfer of electrons between atoms of the elements is called

(a) ionic bond (b) covalent bond
(c) coordinate bond (d) hydrogen bond

77. The compound which contains ionic bond is

(a) CH_4 (b) N_4
(c) CaCl_2 (d) CCl_4

78. Which of the following is a covalent compound?

(a) NaCl (b) CaCl_2
(c) MgCl_2 (d) C_2H_2

79. Strongest bond is

(a) $\text{>C}-\text{C}<$ (b) $\text{>C}=\text{C}<$
(c) $-\text{C}\equiv\text{C}-$ (d) All of these

80. Molten NaCl is a good conductor of electricity due to the presence of

(a) free electrons
(b) free ions
(c) free molecules
(d) None of the above

81. Which one has hydrogen bonding?

(a) HCl (b) HBr
(c) HF (d) HI

82. Hydrogen bonding is maximum in

(a) ethanol (b) diethyl ether
(c) ethyl chloride (d) triethyl amine

- 83.** Salt is soluble in water because
 (a) water forms hydrogen bond with it
 (b) water is a polar solvent and interact with the ions of salt
 (c) water is a non-polar solvent and exists as water of crystallisation
 (d) water loses electrons and form ionic compound with salt
- 84.** In the following reaction,

$$\text{Zn} + [\text{Ag}(\text{CN})_2]^- \longrightarrow [\text{Zn}(\text{CN})_4]^{2-} + \text{Ag}$$

 (a) Zn is reduced and Ag^+ is oxidised
 (b) Zn is oxidised and Ag^+ is reduced
 (c) Zn is reduced and Ag is oxidised
 (d) Neither Zn nor Ag is oxidised
- 85.** Reducing agent can
 (a) accept protons
 (b) donate protons
 (c) accept electrons
 (d) donate electrons
- 86.** Starch iodide paper is used to test for the presence of
 (a) iodine (b) iodide ion
 (c) oxidising agent (d) reducing agent
- 87.** Oxidation number of N in N_3H is
 (a) -3 (b) +3
 (c) $+\frac{1}{3}$ (d) $-\frac{1}{3}$
- 88.** A redox reaction is
 (a) proton transfer reaction
 (b) ion combination reaction
 (c) a reaction in solution
 (d) electron transfer reaction
- 89.** From the following elements, which one has the same oxidation state in all of its compounds?
 (a) Hydrogen (b) Fluorine
 (c) Carbon (d) Oxygen
- 90.** Consider the following statements about ionic compounds.
 1. Ionic compounds are insoluble in alcohol.
 2. Ionic compounds in the solid state are good conductor of electricity.
 Which of these statement(s) is/are correct?
 (a) Only 1 (b) Only 2
 (c) Both 1 and 2 (d) Neither 1 nor 2
- 91.** The fatty substances can be protected from being rancid by
 1. storing them in refrigerator.
 2. keeping them in an inert atmosphere.
 3. adding some antioxidant.
 The correct option is
 (a) 1 and 2 (b) 2 and 3
 (c) 3 and 1 (d) All of these

- 92. Statement I** Nitric acid, hydrogen peroxide and potassium dichromate are reducing agents.
Statement II The substance whose oxidation number increases is oxidised and it is a reducing agent.

Codes

- (a) Both the Statements are correct and Statement II is the correct explanation of Statement I
 (b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
 (c) Statement I is correct but Statement II is incorrect
 (d) Statement I is incorrect but Statement II is correct
- 93.** Consider the following statements
 1. The oxidation state of C in CO_2 is +4 which is its maximum oxidation state.
 2. Nitrous acid (HNO_2) sulphurous acid (H_2SO_3) and sulphur dioxide (SO_2) act as both reducing as well as oxidising agent.

Which of the statement(s) given above is/are correct?

- (a) Only 1
 (b) Only 2
 (c) Both 1 and 2
 (d) Neither 1 nor 2
- 94.** Which of the following statement(s) is/are true?
 1. The process of oxidation leads to a gain of electrons.
 2. The process of oxidation leads to a loss of electrons.
 3. The process of reduction leads to a gain of electrons.
 4. The process of reduction leads to a loss of electrons.

Select the correct answer using the codes given below.

- (a) Only 1 (b) Only 4
 (c) 2 and 3 (d) 1 and 4

- 95.** Match the following Columns

Column I	Column II
A. Ionic bond	1. NH_3
B. Covalent bond	2. Diamond
C. Molecular bond	3. NaCl
D. Hydrogen bond	4. Iodine

Codes

- A B C D
 (a) 1 2 4 3
 (b) 3 2 4 1
 (c) 2 3 1 4
 (d) 4 1 3 2

> Previous Years' Questions

- 96.** $\text{NaOH} + \text{HCl} \longrightarrow \text{NaCl} + \text{H}_2\text{O}$
 In the given chemical reaction, ☞ 2013 II
 (a) sodium is oxidised and oxygen is reduced
 (b) sodium is oxidised and chlorine is reduced
 (c) sodium and hydrogen are oxidised
 (d) None of them are oxidised or reduced
- 97.** Date of manufacture of food items fried in oil should be checked before buying because oils become rancid due to ☞ 2014 I
 (a) oxidation
 (b) reduction
 (c) hydrogenation
 (d) decrease in viscosity

Gas Laws and Solutions

- 98.** At constant temperature, the product of pressure and volume of a given amount of a gas is constant. This is
 (a) Gay-Lussac law
 (b) Charles' law
 (c) Boyle's law
 (d) Avogadro's law
- 99.** Charles' law is represented by
 (a) $V \propto \frac{1}{T}$ (b) $V = T$
 (c) $V \propto T$ (d) $V \propto \frac{1}{T^3}$
- 100.** At constant temperature and pressure, the volume of a gas is directly proportional to the number of molecules. This law had been given by
 (a) Boyle
 (b) Charles
 (c) Graham
 (d) None of the above
- 101.** When a balloon is filled with air, its pressure and volume both increases, so it is a violation of
 (a) Boyle's law (b) Charles' law
 (c) Avogadro's law (d) All of these
- 102.** An ideal gas is one which obeys
 (a) Boyle's law (b) Charles' law
 (c) Avogadro's law (d) All of these
- 103.** In general gas equation $pV = nRT$, V is the volume of
 (a) n moles of a gas
 (b) any amount of a gas
 (c) one mole of a gas
 (d) one gram of a gas

- 104.** Which one of the following is not the value of R ?
 (a) $1.99 \text{ cal K}^{-1} \text{ mol}^{-1}$
 (b) $0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$
 (c) $9.8 \text{ kcal K}^{-1} \text{ mol}^{-1}$
 (d) $8.3 \text{ JK}^{-1} \text{ mol}^{-1}$
- 105.** Cooking of chapatti (roti) is a practical implementation of
 (a) Boyle's law (b) Charles' law
 (c) Avogadro's law (d) All of these
- 106.** Absolute zero is the temperature where all the gases are expected to have
 (a) different volumes
 (b) same volume (c) zero volume
 (d) None of the above
- 107.** The density of the gas is equal to
 (a) np (b) Mp/RT (c) p/RT (d) M/V
- 108.** Deviations from ideal behaviour will be more if the gas is subjected to
 (a) low temperature and high pressure
 (b) high temperature and low pressure
 (c) low temperature
 (d) high temperature
- 109.** The rate of diffusion of hydrogen is about
 (a) one half that of helium
 (b) 1.4 times that of helium
 (c) twice that of helium
 (d) four times that of helium
- 110.** Rate of diffusion of a gas is
 (a) directly proportional to its density
 (b) directly proportional to its molecular mass
 (c) directly proportional to the square of its molecular mass
 (d) inversely proportional to the square root of its molecular mass
- 111.** A bottle of dry ammonia and one of dry hydrogen chloride are connected through a long tube. The stoppers at both ends of the tube are opened, simultaneously. The white ammonium chloride ring will first formed
 (a) at the centre of the tube
 (b) near the hydrogen chloride bottle
 (c) near the ammonia bottle
 (d) throughout the length of the tube
- 112.** A solution which contains the maximum amount of the solute that is dissolved in a given amount of solvent at a particular temperature is called
 (a) saturated solution
 (b) unsaturated solution
 (c) supersaturated solution
 (d) None of the above

- 113.** The number of moles of solute per kg of a solvent is called
 (a) mole fraction (b) normality
 (c) molality (d) molarity
- 114.** 8 g of NaOH is dissolved in 1 L of solution. Its molarity is
 (a) 0.8 M (b) 0.4 M
 (c) 0.2 M (d) 0.1 M
- 115.** The molarity of pure water at 298 K is
 (a) 5.5 M (b) 5.55 M
 (c) 5.50 M (d) 55.55 M
- 116.** In which mode of expression, the concentration of the solution remains independent of temperature?
 (a) Normality (b) Molality
 (c) Molarity (d) Formality
- 117.** At a given temperature, the solubility of a gas in liquid increases with
 (a) increase in temperature
 (b) reduction in gas pressure
 (c) increase in gas pressure
 (d) amount of liquid taken
- 118.** Low concentration of oxygen in the blood and tissues of people living at high altitude is due to
 (a) the low temperature
 (b) the low atmospheric pressure
 (c) the high atmospheric pressure
 (d) both low temperature and high atmospheric pressure
- 119.** Which one of the following is involved for the desalination of sea water?
 (a) Reverse osmosis
 (b) Simple osmosis
 (c) Use of sodium aluminium silicate as zeolite
 (d) Use of ion selective electrodes
- 120.** Scuba divers are at risk due to high concentration of dissolved gases while breathing air at high pressure under water. The tanks used by scuba divers are filled with
 (a) air diluted with helium
 (b) O_2
 (c) N_2
 (d) a mixture of N_2 and helium
- 121.** Which of the following statement(s) is/are correct?
 1. During the pressure of osmosis, the solvent travels from the concentrated solution to the dilute solution.

2. In the reverse osmosis, external pressure is applied to the dilute solution.

Codes

- (a) Only 1
 (b) Only 2
 (c) Both 1 and 2
 (d) Neither 1 nor 2

- 122. Statement I** A chemical reaction becomes faster at higher temperature.

Statement II At high temperature, molecular motion becomes more rapid.

Codes

- (a) Both the Statements are correct and Statement II is the correct explanation of Statement I
 (b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
 (c) Statement I is correct but Statement II is incorrect
 (d) Statement I is incorrect but Statement II is correct

- 123. Match the following Columns**

Column I (Concentration of solution)	Column II (Unit)
A. Molarity	1. mol kg^{-1}
B. Molality	2. mol L^{-1}
C. Normality	3. Gram equiv L^{-1}

Codes

- | | |
|-----------|-----------|
| A B C | A B C |
| (a) 1 2 3 | (b) 3 2 1 |
| (c) 2 1 3 | (d) 2 3 1 |

> Previous Years' Question

- 124.** By what mechanism does scent spread all over the room if the lid is opened? ✎ 2013 II
 (a) Pressure in the bottle
 (b) Compression from the bottle
 (c) Diffusion
 (d) Osmosis

Acids, Bases and Salts

- 125.** An acid is a substance which
 (a) donates a proton
 (b) accepts an electron pair
 (c) gives H^+ in water
 (d) All of the above
- 126.** Formic acid is obtained from
 (a) red ants (b) fats
 (c) vinegar (d) orange
- 127.** An aqueous solution of NH_4Cl is
 (a) basic (b) neutral
 (c) acidic (d) amphoteric

128. Burning of substance *X* produces a gas, solution of which turns red litmus paper blue. The substance *X* is

- (a) base (b) acid
(c) salt (d) None of these

129. Which is not a Lewis base?

- (a) H_2O (b) NH_3
(c) CO_2 (d) BF_3

130. Out of the three solutions, i.e. *A*, *B* and *C*, only one turns blue litmus red. This shows that the remaining two are

- (a) acidic
(b) definitely neutral
(c) either acidic or basic
(d) definitely basic

131. The negative logarithmic value of hydrogen ion concentration is called

- (a) pH (b) pOH (c) $\text{p}K_a$ (d) $\text{p}K_b$

132. Water is neither acidic nor alkaline because

- (a) it boils at a high temperature
(b) it cannot donate or accept electrons
(c) it can dissociate into equal number of hydrogen and hydroxyl ions
(d) it cannot accept or donate protons

133. The pH value of wine is

- (a) 6.5 (b) 2.8 (c) 8.5 (d) 7.0

134. The pH value of sea water is

- (a) 8.5 (b) 2.6 (c) 3.0 (d) 2.5

135. What is the pH value of pure water?

- (a) 1 (b) 6 (c) 7 (d) 10

136. pH values of four solutions *X*, *Y*, *Z*, and *P* are

5, 7, 9 and 2, respectively. Which of these is the strongest acid?

- (a) *X* (b) *Z* (c) *P* (d) *Y*

137. *Aqua-regia* used by chemists to separate silver and gold is a mixture of

- (a) hydrochloric acid (concentrated) and nitric acid (concentrated)
(b) hydrochloric acid (concentrated) and sulphuric acid (concentrated)
(c) nitric acid (concentrated) and sulphuric acid (concentrated)
(d) hydrochloric acid (dilute) and sulphuric acid (dilute)

138. Consider the following statements about pH value

- Human blood has pH value in the range of 7.36-7.42.
- A change in pH of blood by 0.2 blood by 0.2 units results in death.

Which of the statement(s) given above is/are correct?

- (a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

139. When lime juice is dropped on baking soda, brisk effervescence takes place because the gas evolved is

1. hydrogen 2. oxygen
3. carbon dioxide

Choose the correct option.

- (a) Only 1 (b) Only 3
(c) 1 and 3 (d) All of these

140. **Statement I** Solution of blue vitriol (CuSO_4) turns red litmus blue.

Statement II It is a salt of strong acid and weak base.

Codes

- (a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

141. Match the following Columns.

Column I (Acid)	Column II (Use)
A. Boric acid	1. Explosives (TNT)
B. Oxalic acid	2. Preservatives
C. Benzoic acid	3. Rust stain remover
D. Nitric acid	4. Eye wash

Codes

- A B C D
(a) 1 3 2 4
(b) 4 3 2 1
(c) 4 2 3 1
(d) 4 1 2 3

142. **Statement I** Addition of water to an aqueous solution of HCl decreases the pH.

Statement II Addition of water suppresses the ionisation of HCl.

Codes

- (a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

143. Match the following Columns.

Column I (Acid)	Column II (Source)
A. Lactic acid	1. Tamarind
B. Tartaric acid	2. Orange
C. Oxalic acid	3. Tomato
D. Citric acid	4. Sour curd

Codes

- A B C D A B C D
(a) 2 3 1 4 (b) 2 1 3 4
(c) 4 3 1 2 (d) 4 1 3 2

> Previous Years' Questions

144. Which one among the following statements is correct? **☑ 2012 I**

- (a) All bases are alkali
(b) None of the bases is alkali
(c) There are no more bases except the alkalis
(d) All alkalis are bases but all bases are not alkalis

145. The pH of fresh ground water slightly decreases upon exposure to air because **☑ 2012 I**

- (a) carbon dioxide from air is dissolved in the water
(b) oxygen from air is dissolved in the water
(c) the dissolved carbon dioxide of the ground water escapes into air
(d) the dissolved oxygen of the ground water escapes into air

146. Antacids are commonly used to get rid of acidity in the stomach. A commonly used antacid is **☑ 2012 II**

- (a) sodium hydrogen phthalate
(b) magnesium hydroxide
(c) calcium hydroxide
(d) manganese acetate

147. On the labels of the bottles, some soft drinks are claimed to be acidity regulators. They regulate acidity by using **☑ 2012 II**

- (a) carbon dioxide
(b) bicarbonate salts
(c) Both (a) and (b)
(d) carbon dioxide and lime

148. Which of the following solutions will not change the colour of blue litmus paper to red?

1. Acidic solution
2. Basic solution
3. Common salt solution

Select the correct answer using the codes given below **☑ 2014 I**

- (a) 1 and 3 (b) 2 and 3
(c) Only 1 (d) Only 2